Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E Computer Science & Engineering	**Semester**	VI
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Experiment 5

Perceptron vs Multilayer Perceptron with Hyperparameter Tuning

Objective

To implement and compare the performance of:

- **Model A:** Single-Layer Perceptron Learning Algorithm (PLA)
- **Model B:** Multilayer Perceptron (MLP) with hidden layers and nonlinear activation functions

Students must select, tune, and justify hyperparameters such as learning rate, activation function, optimizer, batch size, and number of hidden layers.

Dataset

English Handwritten Characters Dataset

Download Link: https://www.kaggle.com/datasets/dhruvildave/english-handwritten-characters-dataset

- Total samples: 3,410
- Number of classes: 62 (0–9, A–Z, a–z)
- Image type: Grayscale

Theory

Perceptron Learning Algorithm (PLA)

The Perceptron is a linear classifier that uses a step activation function to separate data into two classes.

$$w_{t+1} = w_t + \eta(y - \hat{y})x$$

- Uses a step activation function
- Converges only for linearly separable data

- Extended to multi-class problems using One-vs-Rest strategy

Limitation: PLA cannot learn nonlinear decision boundaries required for handwritten character recognition.

Multilayer Perceptron (MLP)

An MLP consists of one or more hidden layers with nonlinear activation functions.

- Activation functions: ReLU, Sigmoid, Tanh
- Loss function: Categorical Cross-Entropy
- Optimizers: SGD, Adam
- Learns nonlinear decision boundaries using backpropagation

Steps for Implementation

1. Preprocess the dataset (resize, flatten, normalize).
2. Implement **PLA** from scratch:
   - Use step activation function.
   - Apply perceptron weight update rule.
   - Extend PLA to multi-class classification using One-vs-Rest.
3. Implement **MLP**:
   - Select number of hidden layers and neurons.
   - Choose activation functions.
   - Train using backpropagation.
4. Perform **hyperparameter tuning**:
   - Learning rate
   - Batch size
   - Optimizer
   - Activation function
5. Compare PLA and MLP using evaluation metrics.

Performance Evaluation

A/B Experiment Comparison

- **Strengths and Weaknesses of PLA vs MLP:**
  The MLP is better at classifying multiple classes because it can learn nonlinear decision boundaries. In contrast, the PLA is suitable only when the data is linearly separable and may give misleadingly high accuracy on imbalanced datasets by predicting only the majority class.

Table 1: Overall Performance Comparison between PLA and MLP

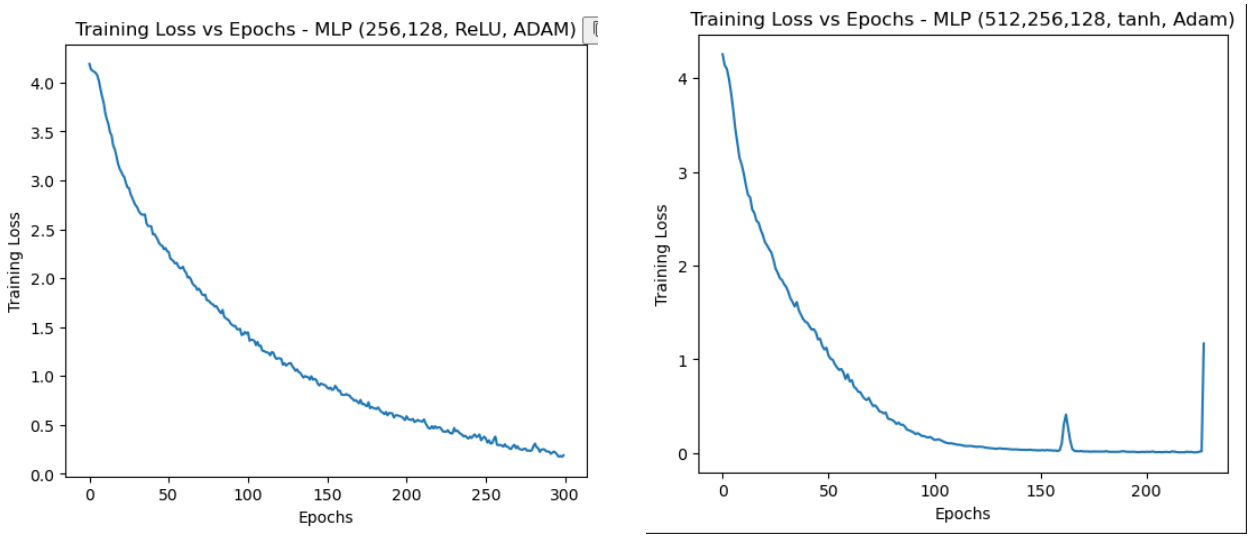
Model	**Accuracy (%)**	**Precision**	**Recall**	**F1-score**
PLA (OvR)	10.11	0.1810	0.1085	0.085
MLP (128)	1.11	0.0003	0.01	0.0002
MLP (256,128)	36.36	0.385	0.36	0.3558
MLP (512,256,128)	23.90	0.33	0.24	0.21

Table 2: Hyperparameter Tuning Results for MLP

Hidden Layers	**Activation**	**Optimizer**	**Learning Rate**	**Batch Size**	**Accuracy (%)**
1 (128)	ReLU	SGD	0.01	32	1.11
2 (256–128)	ReLU	Adam	0.001	64	36.36
3 (512–256–128)	Tanh	Adam	0.0005	64	23.90

Table 3: Training Convergence Comparison

Model	**Epochs**	**Final Training Loss**	**Convergence Behavior**
PLA	100	27	Non-convergent
MLP (Tuned)	228	1.16	Stable convergence

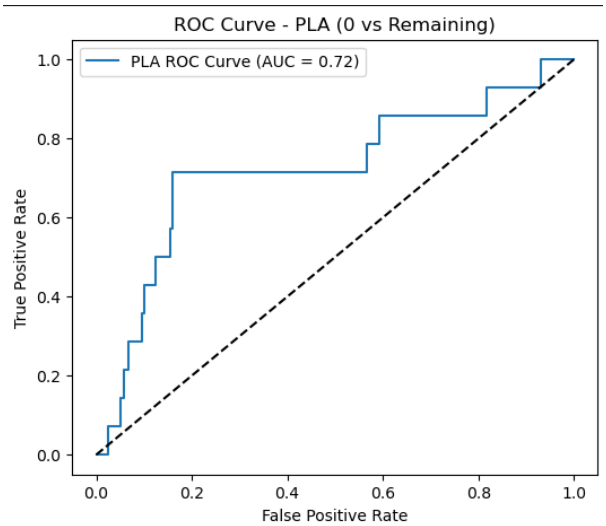


(a) Training Loss - MLP (256,128)

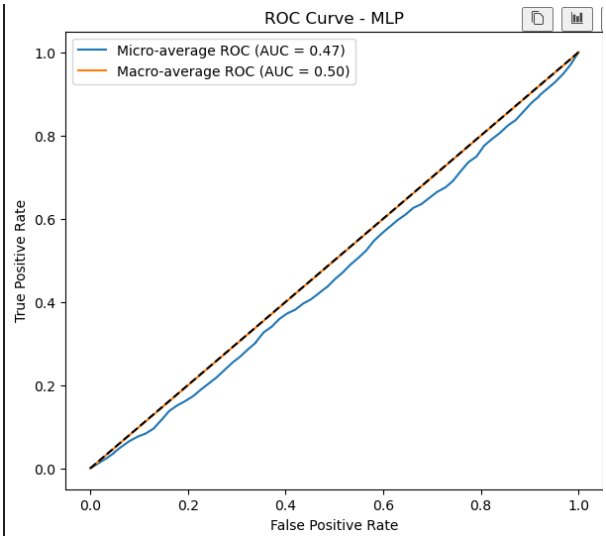
(b) Training Loss - MLP (512,256,128)

Figure 1: Training Loss Comparison for Deeper MLP Architectures

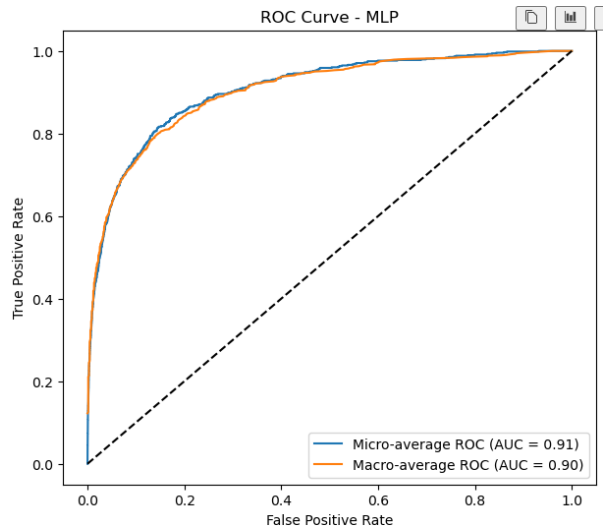
- **Effect of hyperparameter tuning on convergence and accuracy**
  Hyperparamters are generally used to converge in a faster way without leaving out the global



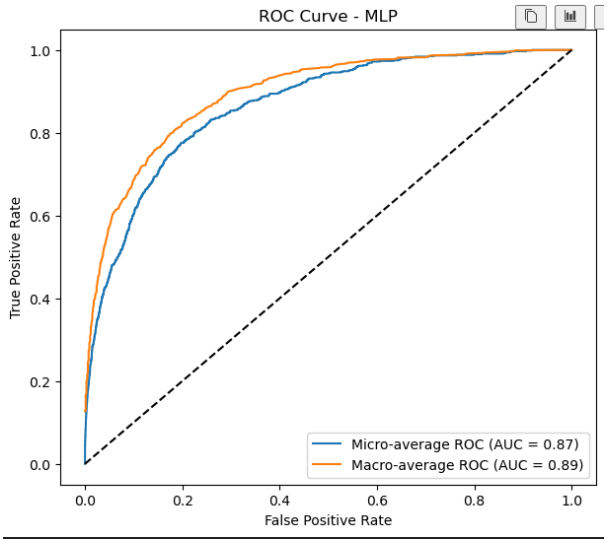
(a) ROC - PLA



(b) ROC - MLP (128)

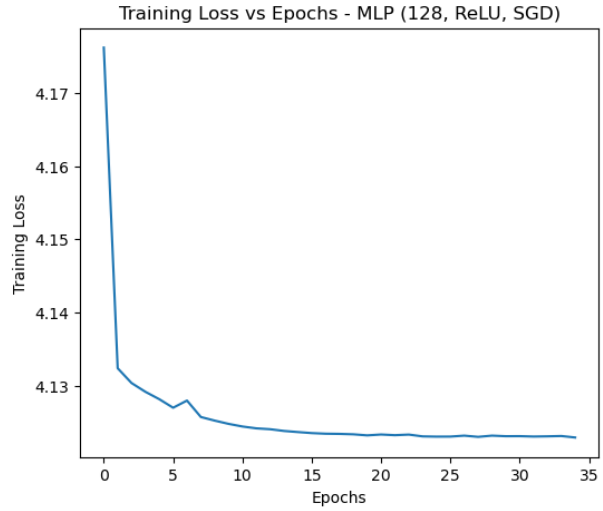


(c) ROC - MLP (256,128)



(d) ROC - MLP (512,256,128)

(e) Training Loss - PLA



(f) Training Loss - MLP (128)

minima and choosing the correct parameter would result Global Minima in a faster way and a good accuracy could be achieved way faster and better than without hyperparamter tuning.

Observation Questions

- **Why does PLA underperform compared to MLP?**
  PLA is a linear classifier and can only learn linearly separable decision boundaries. Handwritten character data is highly nonlinear and high-dimensional, which requires complex feature extraction. MLP uses hidden layers and nonlinear activation functions, enabling it to learn complex patterns and achieve higher accuracy.

- **Which hyperparameter had the most impact on MLP performance?**
  The optimizer and learning rate had the greatest impact on performance. Using Adam with a learning rate of 0.001 resulted in faster convergence and improved accuracy compared to SGD with a lower learning rate.

- **How does optimizer choice affect convergence?**
  SGD updates weights using a fixed learning rate but Adam adapts learning rates leading to faster and more stable convergence.

- **Does increasing hidden layers always improve performance?**
  No. While additional hidden layers increase model capacity and can improve performance, too many layers may cause overfitting, increased computation time, and unstable training.

- **How can overfitting be detected and mitigated?**
  Overfitting is detected when training accuracy is high but validation accuracy is low.